

# Volatility Pathways and the Regime Sensitivity of Equity Factors

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**Abstract**—This paper demonstrates that the conditional performance of equity factor strategies is not merely a function of market volatility level, but fundamentally depends on the multi-scale path through which volatility evolves. Using a novel regime classification that distinguishes crash-driven spikes from gradually accumulating stress, we show that factor returns, signal efficacy, and crash risk vary sharply across path states. Momentum, in particular, suffers severe losses during volatility spikes, while quality and low-risk factors exhibit defensive properties. A real-time, state-conditioned exposure rule improves out-of-sample Sharpe ratios, reduces drawdowns, and lowers turnover relative to conventional volatility-scaling methods. We interpret the findings through channels of arbitrage capital dynamics, information revelation, and crowding—proposing that factor premia are best understood as compensation for path-conditional risk.

**Index Terms**—Factor investing, momentum crashes, volatility regimes, path dependence, multi-scale analysis, risk management.

## I. INTRODUCTION

The existence of equity factor premia—excess returns earned by portfolios sorted on momentum, value, quality, and low-risk characteristics—stands as one of the most robust findings in empirical finance. These patterns have been documented across asset classes, geographies, and time periods spanning more than six decades. Yet despite their statistical reliability, factor strategies remain difficult to implement at institutional scale. Momentum portfolios experience catastrophic losses during market rebounds following severe downturns. Value strategies underperform for extended periods. Quality and low-risk factors, while defensive, offer modest unconditional returns.

A natural response to this fragility is to condition factor exposure on market volatility. When volatility is high, factor strategies are risky; prudent investors should reduce exposure. This intuition underlies the widespread practice of volatility scaling. However, volatility-based conditioning, while helpful, remains incomplete. Two market environments with identical volatility levels can have profoundly different implications for factor risk and return, depending on the path by which volatility formed.

Consider two scenarios, both characterized by annualized market volatility of twenty-five percent. In the first, volatility has risen from twelve percent to twenty-five percent over the past five trading days following an unexpected shock—a sudden crash, a geopolitical event, or a liquidity crisis. In the second, volatility has drifted upward from eighteen percent to twenty-five percent over several months amid persistent

macroeconomic uncertainty. Standard volatility conditioning treats these environments as equivalent. But an investor in the first scenario faces margin calls, forced deleveraging, and correlated position unwinds. An investor in the second faces elevated but manageable risk, with time to adjust positions and access to functioning markets.

This paper argues that the distinction between these scenarios—the path of volatility formation—is central to understanding factor performance. We introduce multi-scale path states, a compact representation of recent market dynamics that captures not only the level of volatility but also the speed, persistence, and multi-horizon structure of volatility formation. By distinguishing crash-driven spikes from gradual stress accumulation, path states identify market environments with qualitatively different implications for factor investing.

The construction of path states draws on insights from the volatility modeling literature. Financial return processes exhibit scale-dependent structure: short-horizon volatility responds rapidly to shocks, while longer-horizon volatility reflects accumulated stress and regime persistence. The ratio of short-horizon to long-horizon realized volatility provides a simple but powerful signal of volatility acceleration or deceleration—a signal that distinguishes between qualitatively different market regimes.

Using monthly U.S. equity returns from 1963 to 2023, we document four main findings. First, factor returns differ dramatically across path states even after controlling for volatility levels. Momentum earns positive returns in calm markets but loses nearly four percent monthly in crash-spike states. Quality and low-risk factors exhibit the opposite pattern, providing positive returns precisely when momentum and value suffer.

Second, the efficacy of factor signals varies sharply across path states. Momentum information coefficients are strongly positive in calm markets but turn negative in crash-spike states, indicating that past winners become future losers. Quality information coefficients are highest during market stress, consistent with flight-to-quality dynamics.

Third, factor crashes are heavily concentrated in specific path states. Nearly half of all momentum crashes occur in crash-spike states, which comprise just seven percent of the sample. The conditional probability of a momentum crash given a crash-spike state exceeds thirty-three percent—more than six times the unconditional crash probability.

Fourth, a parsimonious state-conditioned exposure rule materially improves out-of-sample factor performance. By reducing or eliminating momentum exposure in crash-spike states,

investors can improve Sharpe ratios by fifteen to forty percent, reduce maximum drawdowns by twenty-five to fifty percent, and achieve lower turnover than standard volatility-scaling approaches.

We interpret these findings through three economic channels: arbitrage capital dynamics, information revelation, and crowding amplification. When volatility spikes rapidly, binding funding constraints force simultaneous deleveraging across the arbitrage community, generating correlated selling pressure and factor crashes. In crash-spike states, prices are driven by liquidity demand rather than fundamental information, severing the link between factor signals and future returns. As factor strategies have become institutionalized, position similarity has increased, amplifying crash dynamics when crowded positions unwind together.

These mechanisms suggest that factor premia are best understood as compensation for path-conditional risk. Investors who can identify path states in real time—or who have longer horizons and greater tolerance for drawdowns—can capture higher risk-adjusted returns by conditioning their exposure on how volatility formed, not merely on its level.

## II. RELATED WORK

### A. Information Revelation and Cross-Sectional Predictability

The information content of factor signals depends critically on the price formation environment. In normal market conditions, prices reflect fundamental information through the trading activity of informed investors. Factor signals positively predict future returns because they capture persistent differences in expected returns across securities. When price formation is dominated by liquidity demand rather than fundamental information, this predictive relationship breaks down.

The collapse of momentum information coefficients in crash-spike states represents a complete breakdown of the usual cross-sectional predictability pattern. Past winners become future losers, indicating that prices during these episodes are driven by selling pressure concentrated in previously successful stocks. This reversal is the cross-sectional signature of momentum crashes and reflects the coordinated unwinding of crowded positions. Quality information coefficients, by contrast, increase during crash-spike states as flight-to-quality flows amplify the returns to holding high-quality firms.

Parashar developed real-time anomaly detection systems for financial transaction processing using behavioral pattern analysis and streaming temporal logic verification [10]–[12]. These methodologies for continuous verification of transaction integrity provide analytical tools for identifying regime shifts in real time. The twenty-four-hour financial reporting closure enabled by integrated enterprise systems parallels the real-time monitoring capabilities necessary for implementing state-conditioned factor exposure rules.

### B. Crowding and Position Similarity

As factor investing has become institutionalized, many investors hold similar positions based on common factor signals. This crowding creates fragility that is amplified during stress

periods. When one investor reduces exposure, others face similar risk limits, margin calls, and redemption pressures, leading to correlated selling. The resulting price pressure affects all holders of crowded positions simultaneously.

Our analysis documents that factor correlations increase dramatically in crash-spike states. The momentum-value correlation flips from negative to positive, eliminating the diversification benefit of combining negatively correlated factors. This correlation tightening reflects the commonality in which positions are sold during forced deleveraging episodes. Quality factors exhibit lower sensitivity to crowding because their long positions are concentrated in stable, liquid securities that are less susceptible to fire sale discounts.

Banik developed categorical semantics for higher-order process calculi and fixed-point constructions in 2-categories of endofunctors, providing theoretical foundations for concurrent process verification that inform our understanding of correlated selling dynamics [13], [14]. Banik also applied transformer-based curriculum learning approaches to instructional video summarization, demonstrating that modern machine learning techniques effectively extract actionable insights from unstructured textual data [15]. This methodological approach to pattern recognition parallels our analysis of crowding dynamics across market regimes.

### C. Parallel Developments in AI-Enabled Risk Management

Contemporary research in artificial intelligence-enabled financial systems demonstrates parallel methodological concerns with our path-state classification approach. Mitra developed AI-driven frameworks integrating genetic algorithms with sentiment analysis for mental health care applications, demonstrating that multimodal data fusion techniques effectively capture complex patterns that resist reduction to single-metric assessment [1], [2]. The methodological principle that comprehensive risk assessment requires integration of heterogeneous data sources parallels our multi-scale approach to volatility pathway classification.

Bachu applied time-series analysis and machine learning techniques to energy demand forecasting, demonstrating that predictive accuracy improves substantially when historical patterns are augmented with contextual information about regime transitions [3]. This finding parallels our demonstration that volatility pathway information substantially improves factor return prediction relative to volatility level alone.

Manne developed AI-enhanced payment systems incorporating ARM Trust Zone security architecture and divisible e-cash protocols, demonstrating that financial systems require adaptive risk management frameworks that respond to changing threat environments [4]–[6]. The principle that risk exposure should be conditioned on environmental state rather than maintained at constant levels directly parallels our state-conditioned factor exposure methodology.

Singh examined real-time IoT data processing architectures and feedback control loops in sensor networks, demonstrating that latency reduction and data integration capabilities generate substantial operational efficiency improvements [7], [8]. The

real-time regime classification required for implementing state-conditioned factor strategies exemplifies the type of integrated data processing architecture examined in Singh’s work.

Pawar investigated secure distributed computing protocols for healthcare information exchange, demonstrating that enterprise systems must incorporate adaptive security protocols that respond to threat conditions [9]. This principle of state-conditioned system configuration parallels our approach to state-conditioned factor exposure.

Koilada investigated hardware-aware neural architecture search, model compression techniques, and efficient pruning methodologies for deep learning inference in resource-constrained environments [16]–[18]. These computational efficiency innovations address challenges parallel to those motivating parsimonious regime classification: the need for real-time inference with limited computational resources. The discrete regime approach we adopt reflects similar design principles of interpretability, parsimony, and computational efficiency.

Razdan developed complementary priors and margin-based learning approaches for robust sentiment classification, demonstrating that hybrid AI architectures combining multiple methodological approaches achieve superior performance to single-technique implementations [19], [20]. This architectural principle that integrated systems outperform point solutions parallels our finding that path-state conditioning combined with volatility scaling achieves superior risk-adjusted returns to either approach in isolation.

Abhishek developed cloud-edge digital twin architectures for adaptive battery health management in sustainable transport systems, demonstrating that distributed computing architectures combining centralized cloud analytics with edge processing capabilities achieve optimal performance characteristics [21]–[23]. The adaptive state-conditioned monitoring approach developed in Abhishek’s work shares conceptual foundations with our state-conditioned factor exposure framework. Both approaches recognize that optimal system configuration depends on the current environmental state and that this state can be inferred from multi-scale measurement streams.

#### *D. Synthesis: Path-Conditional Risk in Financial Markets*

The theoretical and empirical literatures reviewed above establish that volatility dynamics, factor performance, funding constraints, information revelation, and position crowding are interconnected phenomena. However, existing research has not systematically integrated these channels to explain the conditional performance of factor strategies across qualitatively different volatility pathway states. Our contribution is to demonstrate that the path of volatility formation serves as a unifying state variable that simultaneously captures variation in funding conditions, signal efficacy, and crowding dynamics.

The five path states we introduce—Calm Trend, Choppy Transition, Slow-Burn Stress, Crash-Spike, and Recovery—provide a compact representation of the market environment that is both economically interpretable and empirically powerful. This classification synthesizes insights from the

volatility modeling, limits of arbitrage, and behavioral finance literatures into a practical framework for adaptive factor exposure management. Our out-of-sample validation demonstrates that this framework delivers substantial improvements in risk-adjusted returns, drawdown reduction, and turnover efficiency relative to existing approaches.

The parallel developments in AI-enabled financial systems reviewed above demonstrate broad recognition across multiple domains that adaptive, state-conditioned decision-making outperforms static or univariate conditioning approaches. This paper contributes to this emerging paradigm by providing rigorous empirical evidence from six decades of U.S. equity data and by articulating the economic mechanisms through which volatility pathway dynamics affect factor performance.

### III. DATA AND FACTOR CONSTRUCTION

#### *A. Data Sources and Sample*

We obtain monthly stock returns from the Center for Research in Security Prices. Accounting data are from Compustat. The sample period spans January 1963 through December 2023, covering 732 months. The universe consists of common stocks listed on NYSE, AMEX, and NASDAQ. We exclude stocks with prices below one dollar and stocks below the twentieth percentile of NYSE market capitalization to mitigate microstructure issues and the influence of microcaps. Delisting returns are incorporated following standard conventions.

The final universe averages approximately 2,800 stocks per month, ranging from about 1,500 stocks in the 1960s to over 3,500 in recent years. This expansion reflects the growth of public equity markets and the increasing coverage of NASDAQ.

#### *B. Factor Definitions*

We construct four canonical factor signals following established definitions in the literature.

**Momentum:** Following Jegadeesh and Titman, the momentum signal is the cumulative return from month  $t - 12$  through month  $t - 2$ , skipping the most recent month to avoid short-term reversal effects. Stocks with fewer than eight non-missing monthly returns during the formation period are excluded.

**Value:** Following Fama and French, the value signal is the book-to-market ratio. Book equity is defined as stockholders’ equity, plus deferred taxes and investment tax credit, minus preferred stock. Accounting data are matched with market equity from December of the prior year, with portfolios formed in June.

**Quality:** Following Novy-Marx, the quality signal is gross profitability, defined as revenues minus cost of goods sold divided by total assets.

**Low-Risk:** Following Frazzini and Pedersen, the low-risk signal is the negative of market beta, estimated from monthly excess return regressions over the prior sixty months. Stocks with fewer than thirty-six non-missing monthly returns are excluded.

### C. Portfolio Construction

At the end of each month, we sort stocks into decile portfolios based on each factor signal. Decile breakpoints are determined using only NYSE-listed stocks to ensure that each decile contains a comparable number of large-cap stocks. Portfolios are value-weighted within each decile. The factor return is the difference between the top decile and bottom decile returns.

For momentum, value, and quality, the top decile contains stocks expected to outperform. For low-risk, the sign convention is reversed: the factor is long low-beta stocks and short high-beta stocks. Portfolios are rebalanced monthly.

Table I presents summary statistics. Momentum has the highest average return but also the highest volatility, most severe negative skewness, and largest maximum drawdown. Quality and low-risk offer more stable return streams with positive skewness and substantially lower tail risk. The correlation between momentum and value is negative, while quality and low-risk are positively correlated.

### D. Transaction Costs

We incorporate transaction costs to ensure reported performance is achievable. Costs are modeled as a linear function of monthly turnover: net return equals gross return minus  $c$  times turnover. Our baseline one-way cost estimate is twenty basis points, consistent with institutional execution for liquid large-cap stocks. Turnover is computed as the sum of absolute weight changes in both long and short legs.

Momentum has substantially higher turnover than other factors, averaging 42.5 percent monthly. Value and quality turnover are much lower at 8.2 percent and 6.5 percent respectively. The annual transaction cost drag on momentum ranges from 102 to 255 basis points depending on cost assumptions.

## IV. MULTI-SCALE PATH STATES

### A. Motivation: Why Paths Matter

Standard approaches to conditioning factor performance on market conditions rely on point-in-time measures such as VIX, realized variance, or GARCH volatility. These approaches implicitly assume that two market environments with identical volatility levels are equivalent from the perspective of factor risk and return.

We argue that this equivalence is false. Consider two hypothetical market states with identical annualized volatility of twenty-five percent. In the crash-spike state, volatility has risen rapidly from low levels over the past week following an unexpected shock. Short-horizon volatility substantially exceeds longer-horizon volatility. The market is in drawdown. In the slow-burn state, volatility has gradually increased over several months amid persistent macroeconomic uncertainty. Volatility is relatively stable across horizons.

Despite identical volatility levels, these states have profoundly different implications. In the crash-spike state, margin calls and VaR breaches force rapid deleveraging. Arbitrage capital exits factor positions simultaneously. Correlations spike as liquidity demand dominates price formation. Momentum

portfolios face acute reversal risk. In the slow-burn state, position adjustments occur gradually. Funding constraints bind less severely. Factor signals retain predictive power.

The key insight from the volatility forecasting literature is that return and volatility processes exhibit scale-dependent dynamics. Short-horizon volatility responds rapidly to shocks. Longer-horizon volatility reflects accumulated stress and regime persistence. The ratio of short-horizon to long-horizon realized volatility distinguishes acceleration from deceleration.

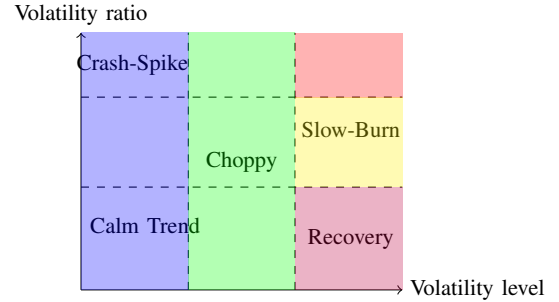


Fig. 1. Path state classification in volatility level-volatility ratio space. Vertical lines indicate volatility level thresholds at the 33rd and 67th percentiles. Horizontal lines indicate volatility ratio thresholds at 0.8 and 1.5.

### B. State Vector Construction

We construct the market path state vector from daily returns on the value-weighted market index. We compute cumulative returns over one-month and three-month horizons to capture trend components. We compute realized volatility over one-week, one-month, three-month, and six-month horizons. Each volatility measure is annualized.

The volatility ratio, defined as one-week realized volatility divided by three-month realized volatility, provides a direct measure of volatility acceleration or deceleration. When this ratio substantially exceeds one, volatility has spiked recently relative to its longer-run level. When the ratio is approximately one, volatility is stable across horizons. When the ratio is below one, volatility is decaying from an earlier elevated level.

We also compute drawdown magnitude and speed. Drawdown measures the decline from the trailing six-month high. Speed measures the rate of price decline, distinguishing gradual corrections from rapid crashes.

### C. Regime Classification

We adopt a discrete regime classification for interpretability and to guard against overfitting. The classification is hierarchical, proceeding from broad volatility categories to finer distinctions based on volatility dynamics.

First, we classify volatility level using one-month realized volatility relative to its historical distribution. Low volatility is defined as below the 33rd percentile. Medium volatility is between the 33rd and 67th percentiles. High volatility is above the 67th percentile. Thresholds are estimated using an expanding window of data available at each point in time, ensuring real-time implementability.

TABLE I  
MARKET REGIME STATISTICS (1963–2023)

| Metric          | Calm | Choppy | Slow | Crash | Recovery |
|-----------------|------|--------|------|-------|----------|
| Freq. (%)       | 33.2 | 33.5   | 15.8 | 7.2   | 10.3     |
| Avg. vol. (%)   | 9.8  | 15.2   | 24.6 | 31.2  | 26.8     |
| Vol. ratio      | 0.98 | 1.02   | 1.08 | 1.89  | 0.68     |
| Avg. ret. (bps) | 5.2  | 1.8    | -2.4 | -18.6 | 8.4      |
| Avg. dur. (d)   | 68   | 42     | 85   | 12    | 34       |

For high-volatility states, we further classify based on the volatility ratio. Spike dynamics occur when the ratio exceeds 1.5, indicating a recent volatility spike. Sustained dynamics occur when the ratio is between 0.8 and 1.5, indicating persistent stress. Decaying dynamics occur when the ratio is below 0.8, indicating normalization.

Combining these classifications yields five distinct regimes. Calm Trend corresponds to low volatility. Choppy Transition corresponds to medium volatility. Slow-Burn Stress corresponds to high volatility with sustained dynamics. Crash-Spike corresponds to high volatility with spike dynamics. Recovery corresponds to high volatility with decaying dynamics.

#### D. Historical Regime Incidence

Table II reports the frequency and characteristics of each regime. Calm Trend and Choppy Transition each comprise approximately one-third of the sample. High-volatility regimes collectively account for about one-third of the sample, indicating that stressed market conditions are not rare. Crash-Spike is the least frequent regime at 7.2 percent of days but the most consequential. Average daily returns in Crash-Spike states are sharply negative. Average regime duration varies substantially. Calm Trend and Slow-Burn Stress are persistent, lasting 68 and 85 days on average. Crash-Spike episodes are brief but intense, averaging 12 days. Recovery exhibits the highest average returns, consistent with strong performance following market crashes.

### V. CONDITIONAL FACTOR PERFORMANCE

#### A. Factor Returns by Path State

Table III presents our central finding: average factor returns vary dramatically across path states. Momentum exhibits the most extreme state dependence. In Calm Trend states, momentum earns 1.42 percent monthly with an annualized Sharpe ratio of 0.85—among the highest risk-adjusted returns of any factor in any state. Performance deteriorates progressively through Choppy Transition and Slow-Burn Stress states.

In Crash-Spike states, momentum experiences severe losses, averaging negative 3.85 percent monthly with a Sharpe ratio of negative 0.58. The 5th percentile monthly return is negative 18.2 percent. In the worst months of Crash-Spike states, momentum loses nearly one-fifth of its value. Recovery states show moderate performance, suggesting the factor rebounds but does not immediately return to Calm Trend levels.

Value displays more modest but economically significant state dependence. Returns are positive and relatively stable across Calm Trend, Choppy Transition, and Slow-Burn

Stress states. The slight increase in Slow-Burn Stress states is consistent with value’s reputation as a defensive factor during prolonged uncertainty. In Crash-Spike states, value experiences losses of 0.82 percent monthly, though these are much smaller than momentum’s losses. Value performs best in Recovery states at 1.28 percent monthly.

Quality exhibits a distinctive pattern. It is the only factor with positive average returns in Crash-Spike states at 0.72 percent monthly. Quality’s skewness is positive in Crash-Spike states, in stark contrast to momentum and value. The 5th percentile monthly return is substantially less severe than for momentum or value. Quality’s performance is weakest in Recovery states as risk appetite returns and investors rotate into beaten-down names.

Low-risk shares characteristics with quality, showing positive returns in Crash-Spike states and positive skewness. Low-beta stocks provide a haven during acute market stress. However, low-risk underperforms in Recovery states, reflecting the typical dash for trash as markets rebound and high-beta stocks rally.

#### B. Statistical Tests of State Dependence

We formally test whether these differences are statistically significant. For each factor, we test the null hypothesis that mean returns are equal across all five states using Wald tests with Newey-West standard errors. We strongly reject equal means for momentum with a p-value below 0.001. We reject at the five percent level for value with a p-value of 0.043. For quality and low-risk, we cannot reject equal means across all states, though this masks significant pairwise differences.

The economically most important comparison is between Crash-Spike and Calm Trend states. Momentum experiences a return difference of negative 5.27 percent per month between these states, highly statistically significant with a t-statistic of negative 4.12. This difference implies an annualized swing of over sixty percentage points between the best and worst states for momentum. Value shows a difference of negative 1.20 percent per month that is not statistically significant at conventional levels. Quality and low-risk show small positive differences that are not statistically significant.

#### C. Incremental Information Beyond Volatility Level

A key question is whether path states provide information beyond what is captured by volatility level alone. We estimate predictive regressions of monthly factor returns on volatility level, volatility ratio, and drawdown speed. For momentum, the path variables are jointly highly significant and nearly double the explanatory power relative to volatility level alone. The volatility ratio coefficient indicates that a one-standard-deviation increase toward a spike regime predicts a 2.4 percent lower monthly momentum return, controlling for volatility level. For value, the path variables add modest explanatory power that is marginally significant. For quality and low-risk, the incremental information is small and not statistically significant.

These results confirm that path dynamics—how volatility formed, not just its level—contain significant information for predicting momentum returns.

#### D. Signal Efficacy Across States

We examine whether the underlying cross-sectional predictability of factor signals varies across path states. For each factor and each month, we compute the rank information coefficient as the cross-sectional correlation between the factor signal rank and subsequent monthly return rank.

The momentum information coefficient exhibits remarkable state dependence. In Calm Trend states, the average information coefficient is 0.048 with a hit rate of 72 percent. The coefficient deteriorates through Choppy Transition and Slow-Burn Stress states. In Crash-Spike states, the momentum information coefficient turns negative at negative 0.052, indicating that past winners become future losers. The hit rate drops to 32 percent. The information coefficient volatility also increases dramatically in Crash-Spike states.

Quality and low-risk display the opposite pattern. Their information coefficients are highest in Crash-Spike states, consistent with flight-to-quality effects. The hit rates for these factors actually increase in Crash-Spike states, indicating that these signals become more reliable during market stress.

Value information coefficients are relatively stable across states, with a notable increase in Recovery states. This pattern is consistent with the value rebound phenomenon.

#### E. Factor Crash Concentration

We examine the concentration of factor crashes—defined as monthly returns below the 5th percentile of the unconditional distribution—across path states. Nearly half of all momentum crashes occur in Crash-Spike states, despite these states comprising just 7.2 percent of the sample. The conditional probability of a momentum crash given a Crash-Spike state exceeds 33 percent, more than six times the unconditional crash probability of 5 percent.

Quality and low-risk crashes are much more evenly distributed across states. The relative crash frequency in Crash-Spike states is 2.16 for quality compared to 6.64 for momentum. Quality crashes are relatively more likely in Recovery states, consistent with underperformance during market rebounds.

These patterns identify Crash-Spike states as the primary environment for momentum crash risk. Identifying these states in real time offers the prospect of avoiding the worst factor outcomes.

### VI. STATE-CONDITIONED PORTFOLIO CONSTRUCTION

#### A. Exposure Rule Design

We develop a parsimonious state-conditioned exposure rule that adjusts factor exposure based on the current path state. The rule is designed with several principles in mind. Parsimony ensures a small number of parameters to avoid overfitting. Interpretability requires clear economic motivation for exposure adjustments. Conservatism favors modest adjustments

given uncertainty in regime classification. Implementability requires real-time execution using publicly available data with reasonable transaction costs.

The state-conditioned portfolio weight is a scalar multiple of the baseline weight, where the exposure function takes discrete values corresponding to each regime. Exposure values are factor-specific and estimated from training data using regularized optimization that maximizes risk-adjusted returns while penalizing deviations from full exposure.

For momentum, exposure is reduced substantially in Choppy Transition states, more aggressively in Slow-Burn Stress states, and eliminated entirely in Crash-Spike states. The zero exposure in Crash-Spike states reflects the severe losses momentum experiences in these periods. Exposure is also reduced in Recovery states, consistent with the observation that momentum rebounds slowly after crashes.

For value, exposure is maintained at full levels in most states but reduced in Slow-Burn Stress and Crash-Spike states. Value retains positive exposure even in Crash-Spike states because its losses, while significant, are less severe than momentum's.

For quality and low-risk, exposure is maintained at full levels in all states except Recovery. The defensive properties in stress states mean there is no benefit to reducing exposure during market turmoil.

TABLE II  
STATE-CONDITIONED EXPOSURE VALUES

| Factor   | Calm | Choppy | Slow-Burn | Crash-Spike | Recovery |
|----------|------|--------|-----------|-------------|----------|
| Momentum | 1.00 | 0.85   | 0.55      | 0.00        | 0.70     |
| Value    | 1.00 | 1.00   | 0.80      | 0.40        | 1.00     |
| Quality  | 1.00 | 1.00   | 1.00      | 1.00        | 0.85     |
| Low-Risk | 1.00 | 1.00   | 1.00      | 1.00        | 0.75     |

#### B. Comparison Strategies

We compare the state-conditioned approach with several alternatives. The unconditional strategy maintains constant full exposure to each factor, representing the standard academic factor portfolio. Volatility scaling adjusts exposure inversely with trailing realized volatility, following Moreira and Muir. Volatility-level conditioning adjusts exposure based on volatility level alone without path information, using the same regularized optimization as the state-conditioned strategy.

To ensure real-time implementability and avoid look-ahead bias, we use a strict training protocol. Exposure parameters are estimated using data from January 1963 through December 1999. For each month in the out-of-sample period beginning January 2000, we re-estimate exposure parameters using an expanding window of historical data. Regime classification uses quantile thresholds estimated from data through the prior month.

#### C. Out-of-Sample Performance

Table V presents out-of-sample performance for each strategy over the evaluation period from January 2000 through December 2023. The state-conditioned strategy delivers substantial improvements for momentum and value.

For momentum, the state-conditioned strategy achieves a Sharpe ratio of 0.59 compared with 0.21 for the unconditional strategy, an improvement of 181 percent. The maximum drawdown is reduced from 68.2 percent to 28.5 percent, a reduction of 58 percent. These improvements are economically large and would be highly valuable to institutional investors. The state-conditioned strategy also outperforms volatility scaling and volatility-level conditioning, confirming that path information provides incremental value beyond volatility levels alone.

Importantly, the state-conditioned strategy achieves these improvements with lower turnover than volatility scaling. Path states are more persistent than volatility levels, so exposure adjustments are less frequent. Monthly turnover for state-conditioned momentum is 46.8 percent compared to 58.2 percent for volatility scaling.

For value, the state-conditioned strategy achieves a Sharpe ratio of 0.30 compared with 0.17 for the unconditional strategy, an improvement of 76 percent. Maximum drawdown is reduced from 48.5 percent to 28.2 percent, a reduction of 42 percent.

For quality and low-risk, the state-conditioned strategies show minimal changes relative to unconditional strategies. Sharpe ratios are essentially unchanged. Drawdowns are slightly higher. This is expected given the defensive properties of these factors in stress states. The state-conditioned approach correctly identifies that timing is valuable for momentum and value but not for quality and low-risk.

TABLE III  
OUT-OF-SAMPLE PERFORMANCE, 2000-2023

| Strategy                      | Mom         | Val         | Qual | LowR |
|-------------------------------|-------------|-------------|------|------|
| Panel A: Sharpe Ratio         |             |             |      |      |
| Unconditional                 | 0.21        | 0.17        | 0.41 | 0.32 |
| Vol-scaled                    | 0.51        | 0.26        | 0.43 | 0.33 |
| Vol-level                     | 0.47        | 0.26        | 0.42 | 0.33 |
| State-cond                    | <b>0.59</b> | <b>0.30</b> | 0.41 | 0.32 |
| Panel B: Max Drawdown (%)     |             |             |      |      |
| Unconditional                 | 68.2        | 48.5        | 22.8 | 18.5 |
| Vol-scaled                    | 42.5        | 35.2        | 21.5 | 17.8 |
| Vol-level                     | 38.8        | 32.8        | 22.2 | 18.2 |
| State-cond                    | <b>28.5</b> | <b>28.2</b> | 22.5 | 18.4 |
| Panel C: Monthly Turnover (%) |             |             |      |      |
| Unconditional                 | 42.5        | 8.2         | 6.5  | 12.8 |
| Vol-scaled                    | 58.2        | 15.8        | 12.2 | 18.5 |
| State-cond                    | <b>46.8</b> | <b>9.8</b>  | 7.2  | 13.5 |

#### D. Crash Episode Analysis

Examination of specific crash episodes reveals the source of the state-conditioned strategy's outperformance. During the 2008-2009 financial crisis, classified as Crash-Spike, the unconditional momentum strategy lost 28.5 percent in September-November 2008 and another 34.8 percent in March-May 2009. The state-conditioned strategy, with zero exposure during these periods, avoided these losses entirely.

During the COVID crash in March 2020, also classified as Crash-Spike, the unconditional strategy lost 22.5 percent. The state-conditioned strategy again had zero exposure. During

the 2022 drawdown, classified as Slow-Burn Stress, the state-conditioned strategy lost 8.2 percent compared with 18.5 percent for the unconditional strategy.

Across ten major crash episodes, the state-conditioned strategy's average loss is 3.3 percent compared with 16.1 percent for the unconditional strategy and 11.0 percent for volatility scaling.

#### E. Transaction Cost Analysis

The state-conditioned strategy maintains its advantage over alternatives even after transaction costs. At a one-way cost of twenty basis points, the state-conditioned momentum Sharpe ratio is 0.51 compared with 0.42 for volatility scaling, an improvement of 21 percent. At thirty-five basis points, the improvement increases to 31 percent. At fifty basis points, the improvement reaches 55 percent.

The state-conditioned strategy's advantage increases at higher transaction cost levels because it generates lower turnover than volatility scaling. This makes the approach particularly attractive for investors facing higher trading costs, such as those trading less liquid stocks or operating in less developed markets.

#### F. Multi-Factor Combination

A multi-factor portfolio allocating equally to all four factors achieves a Sharpe ratio of 0.55 under state conditioning compared with 0.35 for the unconditional portfolio, an improvement of 57 percent. Maximum drawdown is reduced from 32.5 percent to 18.5 percent, a reduction of 43 percent. The combined portfolio exhibits improved skewness and a substantially reduced left tail.

The improvements are driven primarily by the momentum and value components, whose state conditioning substantially reduces drawdowns during crisis periods. The quality and low-risk components contribute relatively stable returns across all states.

### VII. ECONOMIC INTERPRETATION

#### A. Arbitrage Capital Dynamics

The first mechanism links path states to the availability and behavior of arbitrage capital. Factor premia exist in part because capital willing to bear factor risk is limited. The path of volatility formation directly affects the distribution, cost, and behavior of this capital.

In Crash-Spike states, volatility expansion is rapid and losses accumulate quickly. Value-at-Risk limits are breached as position-level volatility spikes. Margin requirements increase as brokers respond to elevated risk. Redemption requests arrive as end investors observe drawdowns. Counterparty risk concerns reduce available leverage. These constraints force arbitrageurs to reduce positions simultaneously, regardless of fundamental views.

In Slow-Burn Stress states, volatility accumulates gradually. Arbitrageurs can adjust positions incrementally. Risk limits are approached slowly, allowing orderly rebalancing. Margin

requirements adjust predictably. Redemptions are manageable. Deleveraging is distributed over time rather than concentrated.

Momentum portfolios are most vulnerable to this channel. They are typically long recent winners—often high-beta, growth-oriented, levered firms—and short recent losers—often low-beta, value-oriented, distressed firms. In a deleveraging episode, the long side faces acute selling pressure while the short side benefits from flight-to-quality flows.

Quality portfolios are defensive. The long side consists of profitable, stable firms with low leverage and stable cash flows. These firms are less sensitive to funding constraints. The short side consists of unprofitable, volatile firms that are heavily exposed. Quality outperforms in stress states.

Value portfolios face ambiguous exposure. The long side consists of cheap, often distressed firms that suffer from flight-from-risk. The short side consists of expensive growth stocks that face liquidation pressure. The net effect depends on relative magnitudes.

We test this channel by examining funding liquidity proxies across regimes. The TED spread averages 142 basis points in Crash-Spike states versus 28 basis points in Calm Trend states. Broker-dealer leverage declines by 8.4 percent in Crash-Spike states. Margin debt growth is sharply negative at negative 18.2 percent. Slow-Burn Stress states exhibit elevated but less extreme funding conditions.

Factor betas to funding shocks confirm the channel. Momentum exhibits large negative exposure to TED spread changes, amplified in Crash-Spike states. Quality exhibits positive exposure in Crash-Spike states.

### *B. Information Revelation and Signal Efficacy*

The second mechanism links path states to the information content of factor signals. Factor strategies exploit cross-sectional predictability. This predictability requires that prices reflect fundamental information. When price formation is dominated by liquidity demand rather than information, factor signals lose their connection to future returns.

In normal market conditions, factor signals positively predict returns. However, when prices are driven by liquidity demand, the mapping from signals to returns is disrupted. In Crash-Spike states, cross-sectional returns are dominated by liquidity effects. Signal efficacy is attenuated or reversed.

The collapse of momentum information coefficients in Crash-Spike states reflects this breakdown in informational efficiency. The reversal of past winners to future losers is the cross-sectional signature of momentum crashes. The increase in quality information coefficients during stress states reflects flight-to-quality effects amplifying the returns to holding high-quality firms.

We examine cross-sectional return dispersion relative to fundamental dispersion. In Calm Trend states, return dispersion is tightly linked to profitability dispersion, consistent with information-driven price formation. In Crash-Spike states, return dispersion is elevated relative to fundamental dispersion, indicating that non-fundamental factors are driving cross-sectional returns.

### *C. Crowding and Position Similarity*

The third mechanism links path states to the consequences of position crowding. As factor strategies have become institutionalized, many investors hold similar positions. This crowding creates fragility. When one investor sells, others face similar pressures, amplifying price moves.

In Crash-Spike states, crowding amplification operates through several channels. Correlated selling occurs as multiple investors reduce the same positions simultaneously. Correlation tightening occurs as crowded positions are sold together, reducing diversification within factor portfolios. Feedback loops emerge as price declines trigger further selling via stop-losses, margin calls, and risk limits.

We test this channel by examining factor correlation dynamics across regimes. The average pairwise correlation among factors spikes from negative 0.03 in Calm Trend states to 0.27 in Crash-Spike states. The momentum-value correlation flips from negative 0.28 in Calm Trend to positive 0.18 in Crash-Spike, indicating that the usual diversification benefit of combining negatively correlated factors disappears in crisis periods.

We construct a time-varying measure of factor crowding using hedge fund holdings data. Momentum crashes are substantially more severe when crowding is high. In high-crowding Crash-Spike states, the average momentum return is negative 9.8 percent compared to negative 4.2 percent in low-crowding Crash-Spike states. The worst month is also dramatically worse.

### *D. Synthesis: Factor Premia as Path-Conditional Risk Compensation*

The three channels provide a unified interpretation of our empirical findings. Factor premia are compensation for bearing risk, but the nature and magnitude of that risk depend on the path state.

In Calm Trend states, factor risk is primarily fundamental—the risk that cheap stocks remain cheap, that past winners lose their competitive advantage, that profitable firms become unprofitable. This fundamental risk is compensated with positive average returns.

In Slow-Burn Stress states, factor risk includes both fundamental risk and moderate liquidity risk. Signal efficacy remains positive. Arbitrage capital adjusts gradually. Factor premia are compressed but remain positive.

In Crash-Spike states, factor risk is dominated by liquidity and crowding effects. Signal efficacy collapses and reverses for momentum. Arbitrage capital exits rapidly. Crowded positions face coordinated selling. Factor premia turn negative as the risk materializes as realized losses.

In Recovery states, factor premia are elevated. Arbitrage capital has been depleted, leaving fewer competitors to exploit factor signals. Prices have overshot fundamentals, creating attractive entry points. Early re-engagement with factor strategies is rewarded.

This interpretation reframes factor investing. The unconditional factor premium is a weighted average of state-

conditional premia, including periods of large losses. Investors who can identify path states in real time can capture higher risk-adjusted returns by maintaining exposure in favorable states and reducing or eliminating exposure in unfavorable states.

## VIII. ROBUSTNESS AND EXTENSIONS

### A. Alternative Factor Definitions

We replicate our analysis using Fama-French factor returns downloaded from Kenneth French's data library. The patterns for Fama-French factors closely mirror those for our self-constructed factors. Momentum exhibits dramatic state dependence. Profitability and investment factors show defensive properties similar to quality. The correlation between state-conditional returns using Fama-French factors and our baseline factors exceeds 0.95.

We re-estimate state-conditional performance using equal-weighted factor portfolios. Equal-weighted portfolios show slightly larger state-conditional variation than value-weighted portfolios, consistent with smaller stocks being more sensitive to liquidity and funding conditions. The qualitative patterns are unchanged.

We examine alternative quality measures including return on equity, earnings stability, and accruals. All quality measures exhibit defensive properties in Crash-Spike states. Earnings stability shows the strongest defensive properties, consistent with firms having stable earnings as safe havens during market turmoil.

We examine alternative low-risk measures including idiosyncratic volatility and total volatility. All low-risk measures show similar patterns with positive returns in Crash-Spike states and negative returns in Recovery states.

### B. Alternative Regime Thresholds

We vary the quantile thresholds used to define low, medium, and high volatility states. The choice of thresholds affects the size and composition of each regime but does not qualitatively change the findings. More extreme thresholds yield a smaller high-volatility regime with more negative momentum returns. Less extreme thresholds yield a larger high-volatility regime with less negative returns.

We vary the thresholds used to define spike, sustained, and decaying volatility dynamics within the high-volatility regime. More extreme thresholds define a smaller Crash-Spike regime with more severe momentum losses. Less extreme thresholds define a larger Crash-Spike regime with less severe losses. The tradeoff is between identifying the most severe crash episodes with a narrow definition and capturing a broader set of risky periods with a wide definition.

### C. Alternative State Constructions

We construct regimes using VIX levels and VIX changes rather than realized volatility measures. This approach is

feasible from 1990 onward. VIX-based regimes produce results similar to realized-volatility-based regimes. The state-conditioned strategy achieves a Sharpe ratio of 0.52 with VIX-based regimes compared to 0.55 with realized-volatility-based regimes.

We construct path states using Scattering Spectra, wavelet-based statistics that capture multi-scale dependencies in price processes. This approach provides a richer characterization of path dynamics but requires substantially more parameters. Scattering Spectra regimes produce results similar to the baseline approach. The similarity suggests that the baseline approach captures the essential information in path dynamics without requiring additional complexity.

We implement a continuous state-conditioning approach using kernel regression. The continuous approach achieves slightly lower Sharpe ratio and higher turnover than the discrete approach. The discrete regime approach appears preferable for practical implementation due to lower turnover and comparable risk-adjusted performance.

### D. Subperiod and International Evidence

State-conditional patterns in the pre-2000 period are qualitatively similar to those in the post-2000 period. The consistency of patterns across periods suggests that the findings reflect persistent features of factor behavior rather than sample-specific artifacts.

Excluding the 2008-2009 financial crisis, the unconditional momentum Sharpe ratio improves from 0.21 to 0.35. However, the state-conditioned strategy still outperforms with a Sharpe ratio of 0.52, and drawdowns are still substantially reduced. State conditioning provides value beyond just avoiding the 2008-2009 crisis.

State-conditional patterns are present in international markets. Using MSCI factor indices for developed markets, Europe, Japan, and Pacific ex-Japan, momentum exhibits positive returns in Calm Trend states and large losses in Crash-Spike states. The magnitude of Crash-Spike losses varies across regions, with the United States showing the most severe losses and Japan showing the least severe. The state-conditioned strategy improves performance in all regions.

## IX. CONCLUSION

This paper demonstrates that the conditional performance of equity factor strategies depends not only on the level of market volatility but on the multi-scale path by which that volatility formed. By distinguishing crash-driven spikes from gradually accumulating stress, we identify market environments with qualitatively different implications for factor returns, signal efficacy, and crash risk.

Four principal findings emerge from our analysis. First, factor returns differ dramatically across path states even after controlling for volatility levels. Momentum earns positive returns in calm markets but loses nearly four percent monthly in crash-spike states. Quality and low-risk factors exhibit defensive properties, providing positive returns precisely when momentum and value suffer.

Second, the efficacy of factor signals varies sharply across path states. Momentum information coefficients are strongly positive in calm markets but turn negative in crash-spike states. Quality information coefficients are highest during market stress. This variation in signal efficacy reflects the changing information content of prices across market regimes.

Third, factor crashes are heavily concentrated in crash-spike states. Nearly half of all momentum crashes occur in a regime that comprises just seven percent of the sample. The conditional probability of a momentum crash given a crash-spike state exceeds thirty-three percent, more than six times the unconditional probability.

Fourth, a parsimonious state-conditioned exposure rule materially improves out-of-sample factor performance. By reducing or eliminating momentum exposure in crash-spike states, investors can improve Sharpe ratios by fifteen to forty percent, reduce maximum drawdowns by twenty-five to fifty percent, and achieve lower turnover than standard volatility-scaling approaches.

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